

Q1 Find smallest and largest number

10 marks

WAP to find smallest and largest number in an array

input: 12 5 18 3 25 9

output:

Smallest number:3

Largest number:25

Your Code

```
1 arr = list(map(int, input().strip().split()))
2 smallest = arr[0]
3 largest = arr[0]
4 for num in arr:
5     if num < smallest:
6         smallest = num
7     if num > largest:
8         largest = num
9 print(f"Smallest number:{smallest}")
10 print(f"Largest number:{largest}")
```

Input

12 5 18 3 25 9

Output

```
Smallest number:3
Largest number:25
```

Q2 Missing Number

10 marks

WAP to find missing number in an array of shuffled order

input: 4 2 1 6 5

output:3

Your Code

```
1 n = int(input())
2 arr=list(map(int,input().split()))
3 total=n*(n+1)//2
4 print(total-sum(arr))
```

Input

6
4 2 1 6 5

Output

3

Q3 Second largest number

10 marks

WAP to find second largest number in an array

INPUT: 12 5 18 3 25 9

OUTPUT: 18

Your Code

```
1 arr=list(map(int, input().split()))
2 arr.sort()
3 print(arr[-2])
```

Input

12 5 18 3 25 9

Output

18

Q4 Most repeated number

10 marks

WAP to print the most repeated number in an array

INPUT: 2 4 6 2 5 3 1 2

OUTPUT: 2

Your Code

```
1 arr=list(map(int,input().split()))
2 max_count=0
3 most_repeated=arr[0]
4 for i in arr:
5     if arr.count(i)>max_count:
6         max_count=arr.count(i)
7     most_repeated=i
8     print(most_repeated)
```

Input

2 4 6 2 5 3 1 2

Output

2
4
6
2
5**Q5 Unique elements**

10 marks

WAP to print all unique elements in an array

INPUT : 2 4 6 2 5 6 7

OUTPUT : 4 5 7

Your Code

```
1 arr=list(map(int,input().split()))
2 for i in arr:
3     if arr.count(i)==1:
4         print(i, end=" ")
```

Input

1 4 7 4 2 2

Output

1 7

Q6 Rotate an array by 'N' positions

10 marks

WAP to rotate an array by 'N' positions

INPUT: 2 5 6 1 3 4

OUTPUT: 6 1 3 4 2 5

Your Code

```
1 arr=list(map(int,input().split()))
2 n=2
3 arr=arr[n:]+arr[:n]
4 print(*arr)
```

Input

1 2 6 4 5 3

Output

6 4 5 3 1 2

Q7 Remove all duplicates in an array

10 marks

WAP to remove all duplicates in an array

INPUT: 2 4 6 2 7 8 3 4

OUTPUT: 2 4 6 7 8 3

Your Code

```
1 arr=list(map(int,input().split()))
2 res=[]
3 for i in arr:
4     if i not in res:
5         res.append(i)
6         print(*res)
```

Input

2 4 6 2 7 8 3 4

Output

```
2
2 4
2 4 6
2 4 6 7
2 4 6 7 8
2 4 6 7 8 3
```

Q8 Print all 0's followed by non-zero without changing the order

10 marks

WAP to print all 0's followed by non-zero without changing the order

INPUT: 1 0 4 0 2 5 0 7 0 6 0

OUTPUT: 0 0 0 0 1 4 2 5 7 6

Your Code

```
1 arr=list(map(int,input().split()))
2 zeros=[]
3 nonzeros=[]
4 for i in arr:
5     if i==0:
6         zeros.append(i)
7     else:
8         nonzeros.append(i)
9     print(*(zeros+nonzeros))
```

Input

1 0 4 0 2 5 0 7 0 6 0

Output

```
Security Error: Disallowed code
pattern detected.
```

Your Code

```
1 s=input()
2 vowels=0
3 consonants=0
4 digits=0
5 spaces=0
6 for ch in s:
7     if ch in "aeiouAEIOU":
8         vowels += 1
9     elif ch.isalpha():
10        consonants += 1
11    elif ch.isdigit():
12        digits += 1
13    elif ch==" ":
14        spaces += 1
15
16    print("vowels:",vowels)
17    print("consonants:",consonants)
18    print("digits:",digits)
19    print("spaces:",spaces)
```

Input

```
vowels:3
consonants:7
digits:3
spaces:2
```

Output

Visible Test Cases

Test Case 1



0.00 / 10 marks

Q10 Palindrome

10 marks

Write a python program to check whether the given string is a palindrome

input: malayalam

output: malayalam is a palindrome

input: saveetha

output: saveetha is not a palindrome

Your Code

```
1 s=input()
2 if s==s[::-1]:
3     print(s,"is a palindrome")
4 else:
5     print(s,"is not a palindrome")
```

Input

malayalam

Output

malayalam is a palindrome